No-Arbitrage Principles and the Fundamental Theorems of Asset Pricing

Lecture 8 Notes — Dominance, Replication, and the Architecture Behind Martingale Pricing

Goals and logical structure

Lecture 7 fixed an equivalent martingale measure $\mathbb{Q}$ and used it to price and hedge. Lecture 8 asks where $\mathbb{Q}$ comes from, and how much can be said before any model is chosen at all:

dominance (one payoff at least as large) $\longrightarrow$ law of one price (identical payoffs)
$\longrightarrow$ replication pricing
$\longrightarrow$ equivalent martingale measures and state prices
$\longrightarrow$ the Fundamental Theorems (existence, uniqueness)
$\longrightarrow$ superhedging bounds when replication fails.

Every inequality or identity below is derived from an explicit trade: what is bought, what is sold, what is the cashflow today, and what is the payoff at maturity. No probability distribution is used until Section 5.

1 A Formal Language for No-Arbitrage

1.1 One-period markets and trading strategies

To state the Fundamental Theorems precisely it helps to work, for a moment, in the simplest possible setting: one period, finitely many states.

Definition 1.1 (One-period market)**.** Fix a finite set of states $\Omega = \{\omega_1, \dots, \omega_K\}$, each with $\mathbb{P}(\omega_k) > 0$, and traded assets $S^0, S^1, \dots, S^m$ with time-0 prices $S_0^0, \dots, S_0^m$ and time-T payoffs $S_T^0(\omega), \dots, S_T^m(\omega)$. Asset S^0 is the money-market account, $S_0^0 = 1$, $S_T^0(\omega) = R > 0$ in every state.

Definition 1.2 (Trading strategy)**.** A trading strategy is a vector $\theta = (\theta^0, \theta^1, \dots, \theta^m) \in \mathbb{R}^{m+1}$, with time-0 value

$$V_0(\theta) = \sum_{i=0}^{m} \theta^i S_0^i,$$

and time-T value

$$V_T(\theta, \omega) = \sum_{i=0}^{m} \theta^i S_T^i(\omega).$$

This is deliberately the discrete skeleton of the self-financing strategies of Lectures 6–7: no rebalancing is needed in one period, so θ is simply a static portfolio.

1.2 Two equivalent notions of arbitrage

Definition 1.3 (Arbitrage)**.** A trading strategy θ is an arbitrage if either

(i) $V_0(\theta) = 0$, $V_T(\theta, \omega) \geq 0$ for every $\omega \in \Omega$, and $V_T(\theta, \omega) > 0$ for at least one ω; or

(ii) $V_0(\theta) < 0$ and $V_T(\theta, \omega) \geq 0$ for every $\omega \in \Omega$.

A market is arbitrage-free if no such θ exists.

Type (i) is a strategy that costs nothing today and sometimes pays off strictly positively; type (ii) is a strategy that is *paid* to enter and never loses money later. Economically these describe the same pathology, and the next proposition makes this precise.

Proposition 1.4 (Equivalence of the two definitions)**.** If a type-(ii) arbitrage exists, then a type-(i) arbitrage also exists.

Proof. Let θ satisfy $V_0(\theta) < 0$ and $V_T(\theta, \cdot) \geq 0$. Augment the strategy with $-V_0(\theta) > 0$ units of the bond S^0, purchased at time 0. The augmented strategy costs

$$V_0(\theta) + \big(-V_0(\theta)\big) = 0$$

today, and its terminal value is $V_T(\theta, \omega)$ plus a strictly positive bond payoff $R \cdot \big(-V_0(\theta)\big) > 0$ in every state, so it is strictly positive everywhere. This is a type-(i) arbitrage. □

Remark 1.5. The converse direction is immediate: a type-(i) arbitrage already has $V_0(\theta) = 0$, which trivially satisfies the weak inequality $V_0(\theta) \leq 0$ used to motivate type (ii). In what follows we use whichever formulation is more convenient for a given argument.

2 The Dominance Principle

2.1 Statement and proof

Proposition 2.1 (Dominance)**.** Let A and B be payoffs at the same maturity T, with current prices a and b. If

$$A \geq B \quad \text{in every state of the world,}$$

then no-arbitrage requires

$$\boxed{a \geq b.}$$

Proof. Suppose instead $a < b$. Buy the claim paying A for a and sell the claim paying B for b. The time-0 cashflow is

$$b - a > 0,$$

and the time-T payoff is

$$A - B \geq 0$$

in every state. This is a type-(ii) arbitrage in the sense of Definition 1.3 with the roles of "buy" and "sell" reversed to make the initial cashflow negative; equivalently, it is exactly the trade constructed there. Contradiction. □

Remark 2.2. Dominance holds contract by contract, state by state. No model, no probability distribution, and no measure enter this argument: it is the most elementary monotonicity statement about prices,

$$\text{larger payoff} \quad \Longrightarrow \quad \text{larger price.}$$

2.2 Applications of dominance

Example — calls are monotone decreasing in the strike

Fix strikes $K_1 < K_2$, same underlying, same maturity. Since

$$(S_T - K_1)^+ \geq (S_T - K_2)^+$$

in every state, dominance gives

$$\boxed{C_t(K_1) \geq C_t(K_2) \quad \text{whenever } K_1 < K_2.}$$

No model of the stock's distribution was used: a lower strike can never be the cheaper call.

Example — default risk cannot make a bond more expensive

A government bond pays \$1 at T for certain. A corporate bond pays \$1 at T if the issuer does not default, and a recovery amount $R_{\text{rec}} < 1$ if it does. The corporate payoff is ≤ 1 in every state, matching the government bond exactly in the no-default state, so dominance gives

$$\boxed{\text{price(corporate bond)} \leq \text{price(government bond)}.}$$

Proposition 2.3 (Convexity of the call price in the strike)**.** Fix $K - h < K < K + h$ with $h > 0$. Then

$$\boxed{C_t(K) \leq \tfrac{1}{2}\big(C_t(K-h) + C_t(K+h)\big).}$$

Proof. Consider the *butterfly* portfolio: long one call at $K - h$, long one call at $K + h$, short two calls at K. Its terminal payoff is

$$f(S_T) = (S_T - K + h)^+ + (S_T - K - h)^+ - 2(S_T - K)^+.$$

Compute f on each region:

- $S_T \leq K - h$: all three terms vanish, $f(S_T) = 0$;
- $K - h < S_T \leq K$: $f(S_T) = (S_T - K + h) - 0 - 0 = S_T - K + h \in [0, h)$;
- $K < S_T \leq K + h$: $f(S_T) = (S_T - K + h) - 0 - 2(S_T - K) = h - (S_T - K) \in (0, h]$;
- $S_T > K + h$: $f(S_T) = (S_T - K + h) + (S_T - K - h) - 2(S_T - K) = 0$.

In every case $f(S_T) \geq 0$, so this payoff dominates the zero payoff. Dominance applied against the claim that pays nothing gives

$$C_t(K - h) + C_t(K + h) - 2C_t(K) \geq 0,$$

which rearranges to the claimed inequality. □

Convexity violations are static, model-free arbitrages

If quoted prices violate $C_t(K) \leq \frac{1}{2}(C_t(K-h) + C_t(K+h))$, the butterfly portfolio above is bought for a strictly negative cost and its payoff is never negative: an explicit arbitrage, independent of any pricing model. Section 9 works a numerical instance of this.

3 The Law of One Price

Corollary 3.1 (Law of one price)**.** If two self-financing strategies with wealth processes X and Y satisfy $X_T = Y_T$ almost surely, then

$$\boxed{X_0 = Y_0.}$$

Proof. Apply dominance in both directions: $X_T \geq Y_T$ gives $X_0 \geq Y_0$, and $Y_T \geq X_T$ gives $Y_0 \geq X_0$. Equivalently: if $X_0 \neq Y_0$, buy the cheaper strategy and sell the more expensive one; the time-0 cashflow is nonzero while the time-T payoff is identically zero, a type-(ii) arbitrage. □

3.1 Replication pricing

Remark 3.2 (The bridge to Lecture 7)**.** If a payoff H is replicated by a self-financing strategy with initial cost X_0, the law of one price forces

$$\boxed{V_0(H) = X_0.}$$

Everything called "martingale pricing" in Lecture 7 is this statement, combined with a convenient way of computing X_0 as a conditional expectation under an equivalent martingale measure. That measure is constructed in Sections 5–7.

Example — put–call parity, rebuilt without a model

At maturity T, with strike K, compare:

Portfolio 1: one call + K zero-coupon bonds, Portfolio 2: one put + one share.

Their terminal payoffs are

$$\text{Portfolio 1: } (S_T - K)^+ + K = \max(S_T, K), \qquad \text{Portfolio 2: } (K - S_T)^+ + S_T = \max(S_T, K).$$

These coincide in every state, so the law of one price gives

$$\boxed{C_t + Ke^{-r(T-t)} = P_t + S_t.}$$

Put–call parity is not a Black–Scholes formula: it is the law of one price applied to a static replication, valid in any arbitrage-free model.

Exercise — the box spread

Fix $K_1 < K_2$ and consider the box

$$\text{long } C(K_1),\ \text{short } C(K_2),\ \text{long } P(K_2),\ \text{short } P(K_1).$$

Compute the terminal payoff for $S_T \le K_1$, for $K_1 < S_T \le K_2$, and for $S_T > K_2$. What identity follows from the law of one price?

Short solution

In every region,

$$(S_T - K_1)^+ - (S_T - K_2)^+ + (K_2 - S_T)^+ - (K_1 - S_T)^+ = K_2 - K_1,$$

a constant. The box replicates a deterministic cashflow, so the law of one price applied against a zero-coupon bond gives

$$\boxed{\big[C_t(K_1) - C_t(K_2)\big] + \big[P_t(K_2) - P_t(K_1)\big] = (K_2 - K_1)e^{-r(T-t)}.}$$

This is put–call parity written at both strikes and subtracted: a model-free identity linking four quoted option prices.

4 Application: Pricing a Forward Contract by Replication

4.1 No dividends

A forward delivering one share at T for a fixed delivery price K_t pays $S_T - K_t$ at maturity. Replicate this payoff statically: buy one share (cost S_t) and sell K_t zero-coupon bonds paying 1 at T (receive $K_t P_t(T)$):

$$\text{cost} = S_t - K_t P_t(T).$$

A forward is entered at zero cost in equilibrium, so the law of one price forces

$$S_t - K_t P_t(T) = 0 \quad \Longrightarrow \quad \boxed{K_t = \frac{S_t}{P_t(T)}.}$$

If the short rate is constant r, then $P_t(T) = e^{-r(T-t)}$ and

$$\boxed{K_t = S_t e^{r(T-t)}.}$$

Example — a one-year forward

With $S_t = 100$, $r = 5\%$, and one year to maturity,

$$K_t = 100\, e^{0.05 \times 1} \approx \boxed{105.13.}$$

No probability, no volatility, no distribution of S_T entered this calculation: the forward price only reflects the cost of carrying the position to T.

4.2 Continuous dividend yield

Exercise — forward price with dividends

Suppose the share pays a continuous dividend yield q. Reinvesting dividends turns $e^{-q(T-t)}$ shares held today into exactly one share at T. Use this prepaid-forward idea, and the no-dividend formula above, to find the delivery price K_t.

Short solution

Buying $e^{-q(T-t)}$ shares today costs $S_t e^{-q(T-t)}$ and delivers exactly one share at T. Replacing S_t by this prepaid cost in the no-dividend formula gives

$$\boxed{K_t = F_{t,T} = S_t e^{(r-q)(T-t)}.}$$

The same replication logic as the no-dividend case, applied to the correct prepaid asset, reproduces the formula used operationally in Lecture 7 without appealing to risk-neutral dynamics.

5 From No-Arbitrage to a Probability Measure

Definition 5.1 (Equivalent martingale measure)**.** A probability measure $\mathbb{Q}$ is an equivalent martingale measure (EMM) for the money-market numeraire if

- $\mathbb{Q} \sim \mathbb{P}$, so $\mathbb{Q}$ and $\mathbb{P}$ have the same null events;
- every discounted traded asset price is a $\mathbb{Q}$-(local) martingale.

This is exactly the role played by $\mathbb{Q}$ in Black–Scholes: under $\mathbb{Q}$, the discounted stock price $D_t S_t$ is a martingale, and Girsanov is the tool that manufactures $\mathbb{Q}$ from $\mathbb{P}$ by changing the drift.

Remark 5.2 (Why equivalence, and not just any probability measure)**.** Suppose a state ω is genuinely possible under $\mathbb{P}$ ($\mathbb{P}(\omega) > 0$), but a candidate measure assigns $Q(\omega) = 0$. A claim paying 1 only in state ω would be priced at 0 under Q, even though it can genuinely cost the seller money. Equivalence is what makes "$\mathbb{Q}$-almost surely" and "$\mathbb{P}$-almost surely" the same statement, so a $\mathbb{Q}$-martingale price is a meaningful economic price and not an artifact of ignoring a real state of the world.

6 The First Fundamental Theorem of Asset Pricing

Theorem 6.1 (FTAP I — informal)**.** A financial market is arbitrage-free if and only if there exists an equivalent martingale measure $\mathbb{Q}$, under which every discounted traded asset price is a $\mathbb{Q}$-(local) martingale:

$$\boxed{\text{no arbitrage} \iff \text{an EMM exists.}}$$

Economic statement	Probabilistic statement
No free lunch from trading	An equivalent martingale measure exists
Prices are internally consistent	Discounted traded gains have no drift under $\mathbb{Q}$

The theorem lets us replace a combinatorial search for arbitrage strategies by an algebraic search for a consistent set of probabilities.

6.1 A binomial illustration

Example — is this one-period market arbitrage-free?

One period, one stock, one bond: $S_0 = 100$, up factor $u = 1.2$, down factor $d = 0.8$, gross risk-free return $R = 1.1$. The martingale condition $pu + (1-p)d = R$ gives

$$p = \frac{R-d}{u-d} = \frac{1.1-0.8}{1.2-0.8} = \boxed{0.75.}$$

Since $0 < p < 1$, a genuine probability exists: the market is arbitrage-free. In general, a valid $p \in (0,1)$ exists if and only if $d < R < u$: the risk-free return must sit strictly between the worst and best stock returns.

Example — when the bond dominates the stock outright

Keep $S_0 = 100$, $u = 1.2$, $d = 0.8$, but now $R = 1.25 > u$, so $d < u < R$ and no valid $p \in (0,1)$ exists. Short one share (receive 100), invest the proceeds at $R = 1.25$:

$$\text{bond payoff} = 125.$$

Buy back the share at $S_T \in \{120, 80\}$:

$$\text{profit} = 125 - S_T \in \{5, 45\}.$$

Strictly positive profit in every state, from a zero-cost trade: exactly the arbitrage that FTAP I predicts must exist whenever no EMM can be found.

6.2 State prices: an equivalent algebraic viewpoint

Definition 6.2 (State prices)**.** In a one-period market with states $\omega_1, \ldots, \omega_K$, a vector $\psi = (\psi_1, \ldots, \psi_K)$ of *state prices* satisfies

$$S_0^i = \sum_{k=1}^{K} \psi_k S_T^i(\omega_k) \qquad \text{for every traded asset } i.$$

Proposition 6.3 (State prices and no-arbitrage)**.** A one-period market with given prices is arbitrage-free if and only if there exists a state-price vector ψ with $\psi_k > 0$ for every k.

This is the same statement as Theorem 6.1, in different units: setting $p_k = \psi_k R$ turns a positive state-price vector into a genuine risk-neutral probability vector, and conversely.

Example — state prices from a bond and a stock

Bond paying 1 in both states, priced at $B_0 = 1/1.1 \approx 0.9091$; stock with $S_0 = 100$, payoffs 120 (up) and 90 (down). Solve

$$\psi_1 + \psi_2 = 0.9091, \qquad 120\psi_1 + 90\psi_2 = 100$$

to get

$$\boxed{\psi_1 \approx 0.6061, \quad \psi_2 \approx 0.3030.}$$

Both are positive, so this price system is arbitrage-free. Rescaling by the bond return recovers the familiar risk-neutral probability:

$$p = \psi_1 R = 0.6061 \times 1.1 \approx \tfrac{2}{3}, \qquad \tfrac{2}{3}(120) + \tfrac{1}{3}(90) = 110 = 100 \times 1.1.$$

Example — a negative state price signals an arbitrage

Keep the same bond and stock payoffs, but now suppose the stock is quoted at $S_0 = 115$. Solving

$$\psi_1 + \psi_2 = 0.9091, \qquad 120\psi_1 + 90\psi_2 = 115$$

gives

$$\psi_1 \approx 1.106, \qquad \psi_2 \approx -0.197.$$

No positive solution exists: $S_0 = 115$ is too high relative to the payoffs and the bond.

reading a negative state price

FTAP I in one line: existence of an arbitrage-free price system is exactly existence of a strictly positive solution to the pricing equations. A negative "price" is the algebraic signature of an arbitrage, and it translates directly into a trade: sell the overpriced stock, and hold the bond and cash instead.

7 The Second Fundamental Theorem of Asset Pricing

Theorem 7.1 (FTAP II — informal)**.** In an arbitrage-free market, the market is complete (every claim is replicable) if and only if the equivalent martingale measure is unique:

$$\boxed{\text{market complete} \iff \text{EMM unique.}}$$

- Unique EMM $\Longrightarrow$ every replicable claim has one unambiguous price.
- Several EMMs $\Longrightarrow$ no-arbitrage alone leaves a range of admissible prices for unreplicable claims.

Example — one bond and one stock cannot pin down three states

One period, one bond ($R = 1.1$), one stock with $S_0 = 100$ and three possible terminal values $S_T \in \{130, 100, 70\}$ (states u, m, d). A risk-neutral probability must satisfy

$$p_u + p_m + p_d = 1, \qquad 1.3p_u + 1.0p_m + 0.7p_d = 1.1,$$

two equations for three unknowns. Writing $p_m = t$,

$$p_u = \tfrac{2}{3} - \tfrac{1}{2}t, \qquad p_d = \tfrac{1}{3} - \tfrac{1}{2}t, \qquad t \in \left(0, \tfrac{2}{3}\right)$$

gives infinitely many valid probability vectors. Two traded assets cannot span three states: exactly the FTAP II statement that this market is incomplete, so the EMM is not unique.

Example — different EMMs price an unreplicable claim differently

Take $t = 0.2$ and $t = 0.4$ in the family above:

$$t = 0.2: \ (p_u, p_m, p_d) = (0.5667,\ 0.2,\ 0.2333), \qquad t = 0.4: \ (0.4667,\ 0.4,\ 0.1333).$$

Consider a claim paying 1 only in the middle state m. Its price is p_m/R under each measure:

$$\frac{0.2}{1.1} \approx 0.182, \qquad \frac{0.4}{1.1} \approx 0.364.$$

Two legitimate EMMs, two different prices: no-arbitrage alone cannot select one.

Example — a third traded asset restores completeness

Suppose the market also quotes a call struck at 100 on the same stock, with payoffs $(30, 0, 0)$ in states (u, m, d) and observed price $C_0 \approx 13.64$. Its pricing equation depends only on p_u:

$$\frac{30\,p_u}{1.1} = 13.64 \quad \Longrightarrow \quad p_u = 0.5.$$

Substituting into $p_u = \frac{2}{3} - \frac{1}{2}t = 0.5$ gives $t = \frac{1}{3}$, hence

$$\boxed{(p_u, p_m, p_d) = \left(\tfrac{1}{2},\ \tfrac{1}{3},\ \tfrac{1}{6}\right),}$$

a single, uniquely determined EMM. The payoff vectors of the bond $(1, 1, 1)$, the stock $(1.3, 1.0, 0.7)$, and the call $(30, 0, 0)$ are linearly independent: three genuinely different exposures for three states.

Remark 7.2. Completeness is about spanning payoffs, not about counting traded assets. This is literally why traders add options to a hedging book: a derivative with a payoff the existing assets cannot replicate expands the span of attainable claims, exactly as FTAP II predicts.

8 Superhedging and Price Bounds

Definition 8.1 (Superhedge and subhedge)**.** A self-financing strategy with wealth X *superhedges* a claim H if $X_T \geq H$ almost surely. A strategy with wealth Y *subhedges* H if $Y_T \leq H$ almost surely.

By dominance,

$$\boxed{Y_0 \leq V_0(H) \leq X_0}$$

for every subhedge Y and superhedge X. When $X_T = H$ exactly, the upper and lower bounds coincide and $V_0(H) = X_0$: exact replication is the special case in which superhedging and subhedging meet.

Example — call bounds from parity and positivity

From Section 3, $C_t - P_t = S_t - Ke^{-r(T-t)}$. Combine this with $P_t \geq 0$ and $(S_T - K)^+ \leq S_T$ to get, with no distributional assumption at all,

$$\boxed{\left(S_t - Ke^{-r(T-t)}\right)^+ \leq C_t \leq S_t.}$$

Exercise — bounding a best-of-two-stocks claim

Let $H = \max(S_T^1, S_T^2)$. Find a subhedging portfolio using a single stock, and a superhedging portfolio using both stocks. What price bounds follow?

Short solution

Since $\max(a, b) \geq a$ and $\max(a, b) \geq b$, holding stock i alone subhedges:

$$S_t^i \leq V_t(H) \quad \text{for } i = 1, 2 \quad \Longrightarrow \quad \max(S_t^1, S_t^2) \leq V_t(H).$$

Since $\max(a, b) \leq a + b$ for $a, b \geq 0$, holding both stocks superhedges: $V_t(H) \leq S_t^1 + S_t^2$. Hence

$$\boxed{\max(S_t^1, S_t^2) \leq V_t(H) \leq S_t^1 + S_t^2.}$$

Two static portfolios, zero modeling assumptions, a genuine price corridor.

8.1 Bounding a digital option with call spreads

A digital pays $H = \mathbf{1}_{\{S_T > K\}}$. Only the market's call prices $C_t(\cdot)$ are used, no closed form.

For small $h > 0$, consider the scaled bull spread

$$\text{spread}(S_T) = \frac{(S_T - K)^+ - (S_T - (K+h))^+}{h} = \begin{cases} 0, & S_T \leq K, \\ (S_T - K)/h, & K < S_T < K + h, \\ 1, & S_T \geq K + h. \end{cases}$$

On $(K, K+h)$ the spread is strictly below 1 while the digital already equals 1; elsewhere the two agree. So the spread never exceeds the digital, and dominance gives the lower bound

$$\boxed{\frac{C_t(K) - C_t(K+h)}{h} \leq \text{price}\left(\mathbf{1}_{\{S_T > K\}}\right).}$$

Exercise — the matching superhedge

Repeat the argument with the bull spread struck at $K - h$ and K instead of K and $K + h$. What inequality do you get?

Short solution

The spread $\big[(S_T-(K-h))^+-(S_T-K)^+\big]/h$ dominates the digital everywhere by the symmetric argument, giving the upper bound. Combining both,

$$\boxed{\frac{C_t(K)-C_t(K+h)}{h} \le \text{price}\big(\mathbf{1}_{\{S_T>K\}}\big) \le \frac{C_t(K-h)-C_t(K)}{h}.}$$

As $h \to 0$, both bounds converge to $-\partial_K C_t(t,K)$, the classical model-free relationship between digital prices and the slope of the call price curve in the strike.

9 Guided Practice and Final Synthesis

Exercise — diagnose an arbitrage from raw quotes

Same maturity, same stock, three quoted calls:

$$C_t(90)=14, \qquad C_t(100)=12, \qquad C_t(110)=7.$$

Check monotonicity in the strike, then check convexity in the strike. If a principle from this lecture is violated, construct the explicit trade and its guaranteed profit.

Short solution

Monotonicity: $14>12>7$, consistent with dominance. Convexity: the butterfly bound requires

$$C_t(100) \le \tfrac{1}{2}\big(C_t(90)+C_t(110)\big) = \tfrac{1}{2}(14+7) = 10.5,$$

but the quote is $12>10.5$: the middle strike is overpriced relative to the wings. Buy $C_t(90)$ and $C_t(110)$, sell two units of $C_t(100)$:

$$\text{cost} = 14+7-2(12) = -3,$$

a receipt of \$3 today, with terminal payoff the butterfly tent, which is ≥ 0 in every state. Cash today, no possible loss at maturity: exactly the type-(ii) arbitrage from Definition 1.3.

Four ideas, stacked in order

1. **No-arbitrage principles** (dominance, law of one price) rule out inconsistent prices, with no model.
2. **Replication** gives an exact price whenever a payoff can be synthesized from traded assets.
3. **Equivalent martingale measures** turn no-arbitrage into a pricing rule.
4. **Completeness** determines whether that computation returns one price or only a range.

Five checks before trusting a claimed no-arbitrage price

1. Is the comparison payoff actually dominating, in *every* state?
2. Is the replicating strategy genuinely self-financing?
3. Is the candidate pricing measure equivalent to $\mathbb{P}$, not merely a probability measure?
4. Is the EMM unique for this market, or only for the traded assets?
5. If replication fails, have you produced *both* a superhedge and a subhedge, not just one?

Final synthesis

$$\boxed{\text{no-arbitrage} \iff \text{martingale pricing under a suitable measure.}}$$

Lecture 7 *used* an EMM to price and hedge; Lecture 8 explains its existence (FTAP I), its uniqueness (FTAP II), and the bounds available without replication. Dominance, replication, and the FTAP are the same idea at four levels of abstraction.